

Tzu-Wei Wang (William)

Email: william921127@gmail.com | Phone: (+65) 8426 4882 | LinkedIn: [linkedin.com/in/tzu-wei-wang](https://www.linkedin.com/in/tzu-wei-wang)

RESEARCH FOCUS

Quantitative electron microscopy and data-driven materials research, with emphasis on 4D-STEM, electron ptychography, orientation and strain mapping, and microstructure–property relationships.

EDUCATION

Nanyang Technological University (NTU), Singapore

Expected May 2027

Bachelor of Engineering in Materials Science and Engineering & Data Analytics (Double Major)

- GPA: 4.84/5.00 (Dean's Distinction Award, AY2023/24)

RESEARCH EXPERIENCE

Incoming Research Intern, National Institute for Materials Science (NIMS), Japan

Dec 2026 – Jun 2027

- Will investigate deuterium-induced nanoscale changes in BCC refractory HEAs for hydrogen storage using APT and TEM.

Final Year Project Researcher, Nanyang Technological University, Singapore

Jul 2026 – Present

- Develop 4D-STEM workflows for orientation/strain mapping and ptychography (PtyRAD) on Merlin and EPAD datasets.
- Diagnosed thickness-limited ptychography in a refractory HEA & MEA; benchmarked NBED and GPA for strain mapping.
- Validated the ptychographic reconstruction pipeline on a public WSe₂ dataset.

Research Intern, Emerging nanoscience Research Institute (EnRI), Singapore

May – Jul 2026

- Characterized microstructure of 3 HEA-based structural energetic materials using XRD and SEM.
- Quantified energy release and mechanical response via DSC/TGA, bomb calorimetry, compression, and nanoindentation.

Research Intern, Forschungszentrum Jülich, Germany

Jun – Aug 2025

- Simulated TEM images with Dr. Probe to distinguish true microstructural features from imaging artifacts.

Undergraduate Researcher, URECA Programme, NTU, Singapore

May 2025 – Jun 2026

- Curated 907 battery-material records into 171 unique compositions for reproducible ML screening of Li-ion cathodes.
- Trained CatBoost models reaching $R^2 = 0.705$ (record-level) and 0.527 (unseen-composition holdout).

Undergraduate Researcher, URECA Programme, NTU, Singapore

Oct 2023 – Mar 2025

- Measured activity and selectivity of transition-metal electrocatalysts for H₂O₂ electrosynthesis via CV, LSV, and RRDE.

CONFERENCE PRESENTATIONS

Wang, T.-W., & Millner, G. (2026). "Quantitative Microscopy and Informatics for Property Prediction in Commercial Stainless Steels". Abstract accepted to Microscopy & Microanalysis (M&M 2026), Milwaukee, WI, USA

Wang, T.-W., & Millner, G. (2026). "Explainable inverse design of battery materials via multi-model learning and conditional filtering". Presented at The Minerals, Metals & Materials Society (TMS 2026) Annual Meeting, San Diego, CA, USA.

Wang, T.-W., & Zengyu, H. (2025). "Machine learning assisted mapping research of electrocatalysts study for hydrogen peroxide electrosynthesis". Oral & poster presentation, International Conference of Undergraduate Research (ICUR 2025), Singapore.

PUBLICATIONS & RESEARCH OUTPUTS

Wang, T.-W.; Millner, G.; Wang, Z.; Zhou, W.; Chen, E. Y.-S. (2026). "Composition-Level Prioritization of Lithium-Ion Cathode Candidates." *Under review at cScience*.

Millner, G.; **Wang, T.-W.**; Shen, S.; Li, Y.; Chen, E. Y.-S. (2026). "A Transferable, Explainable AI Framework for Composition–Structure–Property Mapping from Steel Micrographs." *Under review at Materials Today Communications*.

Chou, C.-Y.; Zeng, X.-Y.; Shen, S.; **Wang, T.-W.**; Chen, E. Y.-S.; Juang, J.-Y. (2026). "Enhanced carrier transport in H-doped ZnO films via one-step atmospheric pressure plasma jet." *Submitted to the Journal of Alloys and Compounds*.

Millner, G., **Wang, T.-W.** (2025). "Explainable AI for Property Prediction in Scrap-Based Steel Production: A Public Data Implementation." Conference paper and oral presentation, European Steel Technology and Application Days (ESTAD 2025), Verona, Italy. Related data/supplementary materials published in DR-NTU Data. DOI: [10.21979/N9/NYRGAA](https://doi.org/10.21979/N9/NYRGAA)

Manuscripts in Preparation

Wang, T.-W., et al. (2026). *“A 4D-STEM Ptychography Workflow for Multiscale Strain Mapping”*

Wang, T.-W., et al. (2026). *“Phase Formation and Reactivity in Refractory High-Entropy Alloys”*

Jing, J., **Wang, T.-W.**, et al. (2026). *“Irradiation-Induced Structural Evolution of TiC Across Temperatures”*

Zengyu, H., **Wang, T.-W.**, et al. (2025). *“Machine learning assisted mapping research of electrocatalysts study for hydrogen peroxide electrosynthesis”*

AWARDS & FUNDING

1 st Place, URECA Poster Competition, College of Engineering, NTU (Sole invited speaker)	2026
NTU President Research Scholar, URECA Programme	2025
3 rd Place, URECA Poster Competition, College of Engineering, NTU	2025
Students’ Global Experience (SGE) Award, NTU — USD 1,560	2025
1 st Runner-Up, SimplifyNext RPA x Gen AI Hackathon	2024
Gold Medal, International Genetically Engineered Machine (iGEM) Competition	2023
International Research Funding Program (MOE, Taiwan) — USD 7,500	2023
National Tsing Hua University Research Grant — USD 7,190	2023
Amway Hope Maker “Dream Big, Aim High” Dream Pursuit Grant — USD 9,460	2023
Zhu-Meng (Dream-Building) Scholarship, National Tsing Hua University — USD 5,000	2023

PROFESSIONAL & LEADERSHIP

Vice President, ACS Student Chapter (NTU)	Aug 2025 – Jun 2026
- Led 5 student events; chapter won the 2025–2026 ACS Commendable Student Communities Award.	
Trainee, Graduate Research Innovation Programme (GRIP) Phase I, NTU	Jan – May 2025
- Interviewed 56 prospective customers and delivered 3 pitches on commercializing a research-based technology.	

SKILLS

Electron Microscopy: 4D-STEM, TEM/STEM, SEM, EBSD, FIB, ion milling

Materials Characterization: XRD, DSC/TGA, bomb calorimetry, nanoindentation, CV/LSV/RRDE

Computational & Programming: PtyRAD, Dr. Probe, GPA, UMLIP, Python (PyTorch, scikit-learn, CatBoost), Git, LaTeX